Intelligence artificielle : enjeux et outils

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Course outline

- What is artificial intelligence?
- Machine learning
- Data quality
- Deep learning
- Generative AI
- Societal challenges of AI

What is artificial intelligence?

The literal meaning of artificial intelligence

1. Intelligence

The set of mental functions concerned with conceptual and relational knowledge.

- Larousse

2. Artificial

Produced by human activity, as opposed to nature.

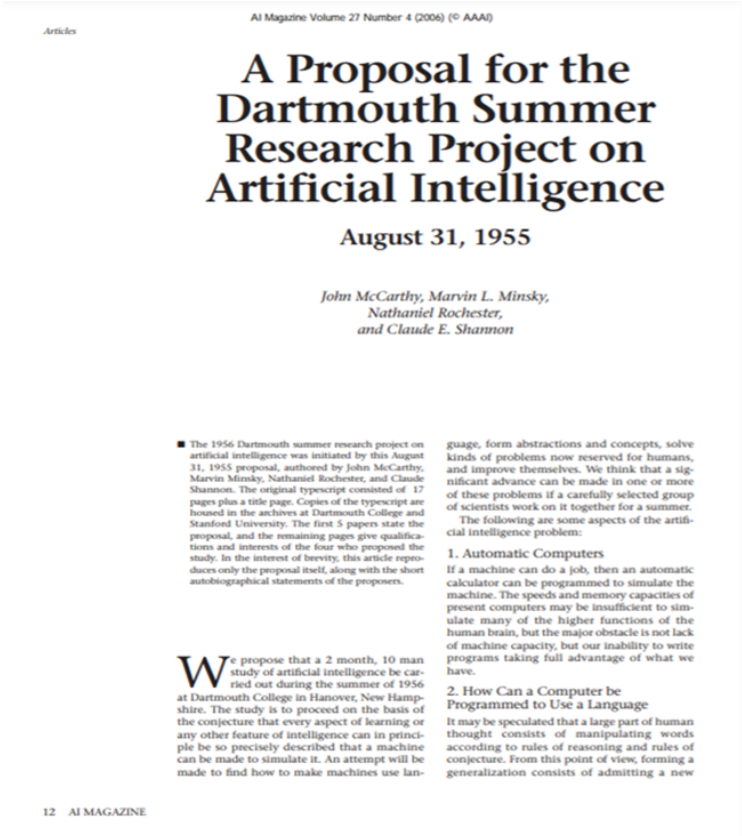
- Larousse

What is intelligence?

Intelligence encompasses several cognitive abilities:

1. **Reasoning:** The ability to solve problems and make logical deductions.
2. **Learning:** The ability to acquire new knowledge and improve through experience.
3. **Perception:** The ability to recognize and interpret sensory stimuli.
4. **Understanding:** The ability to grasp the meaning and significance of concepts and situations.
5. **Memory:** The ability to store and recall information.
6. **Creativity:** The ability to invent or produce new ideas and think in original ways.

The Dartmouth Conference?



Articles

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A Proposal for the Dartmouth Summer Research Project on Artificial Intelligence

August 31, 1955

John McCarthy, Marvin L. Minsky, Nathaniel Rochester, and Claude E. Shannon

■ The 1956 Dartmouth summer research project on artificial intelligence was initiated by this August 31, 1955 proposal, authored by John McCarthy, Marvin Minsky, Nathaniel Rochester, and Claude Shannon. The original typescript consisted of 17 pages plus a title page. Copies of the typescript are housed in the archives at Dartmouth College and Stanford University. The first 5 papers state the proposal, and the remaining pages give qualifications and interests of the four who proposed the study. In the interest of brevity, this article reproduces only the proposal itself, along with the short autobiographical statements of the proposers.

We propose that a 2 month, 10 man study of artificial intelligence be carried out during the summer of 1956 at Dartmouth College in Hanover, New Hampshire. The study is to proceed on the basis of the conjecture that every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it. An attempt will be made to find how to make machines use language, form abstractions and concepts, solve kinds of problems now reserved for humans, and improve themselves. We think that a significant advance can be made in one or more of these problems if a carefully selected group of scientists work on it together for a summer.

The following are some aspects of the artificial intelligence problem:

1. Automatic Computers

If a machine can do a job, then an automatic calculator can be programmed to simulate the machine. The speeds and memory capacities of present computers may be insufficient to simulate many of the higher functions of the human brain, but the major obstacle is not lack of machine capacity, but our inability to write programs taking full advantage of what we have.

2. How Can a Computer be Programmed to Use a Language

It may be speculated that a large part of human thought consists of manipulating words according to rules of reasoning and rules of conjecture. From this point of view, forming a generalization consists of admitting a new

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"We propose that a 2-month, 10-man study of artificial intelligence be carried out during the summer of 1956 at Dartmouth College in Hanover, New Hampshire. *The study is to proceed on the basis of the conjecture that every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it. An attempt will be made to find how to make machines use language, form abstractions and concepts, solve kinds of problems now reserved for humans, and improve themselves. We think that a significant advance can be made in one or more of these problems if a carefully selected group of scientists work on it together for a summer."*

Artificial intelligence according to John McCarthy

“It is the science and engineering of making intelligent machines, especially intelligent computer programs. It is related to the similar task of using computers to understand human intelligence, but AI does not have to confine itself to methods that are biologically observable.”

- John McCarthy

The Turing Test

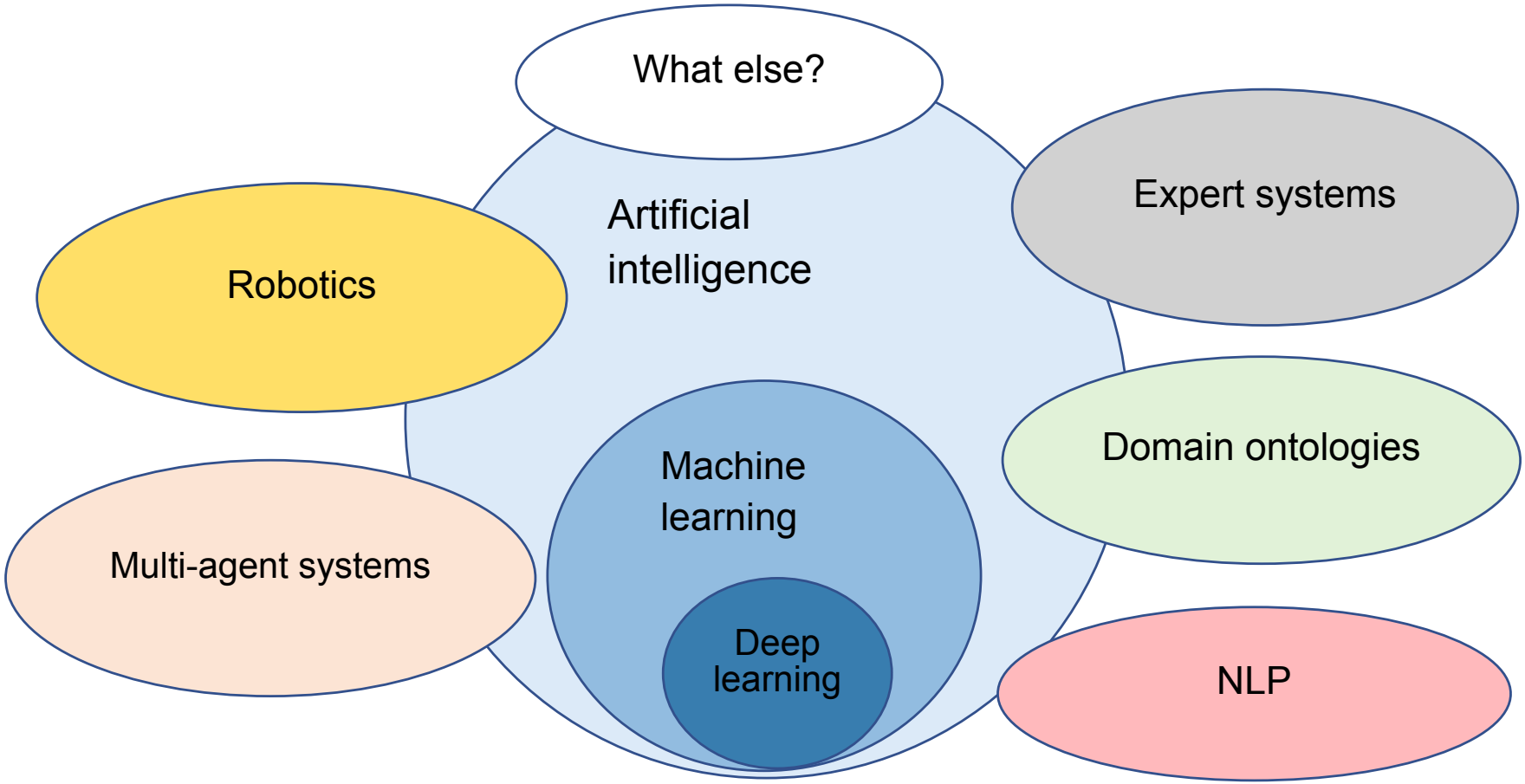
Developed by Alan Turing in 1950, the Turing Test attempts to assess a machine's intelligence, or more precisely its ability to think. Because this is difficult to measure, the test substitutes the ability to process natural language like a human for the ability to think.

Key features of the Turing Test:

- A human interrogator converses with a human and a machine, both hidden from view.
- If the interrogator cannot consistently identify the machine, it is considered to have passed the test.
- The test does not measure knowledge or truthfulness, but the ability to reproduce human behaviour.

So, what is artificial intelligence?

A set of techniques that enable machines to perform tasks traditionally requiring human intelligence.



Two views of artificial intelligence

Symbolic AI (GOFAI)

- Rooted in logic and rationalist philosophy
- Explicit knowledge representation
- Rule-based reasoning and logical inference
- Used in expert systems

“Thinking means manipulating symbols according to rules.”

Connectionist AI

- Inspired by the human brain
- Distributed, implicit representation
- Learning by adjusting synaptic weights
- The foundation of deep learning

“Thinking means learning through interaction with the world.”

Two worldviews, two ways of conceiving intelligence: logic versus emergence.

Connectionist AI vs symbolic AI

Symbolic artificial intelligence: Systems based on rules and symbols to imitate human reasoning.

- Logic
- Sets of rules
- Knowledge-driven

Connectionist artificial intelligence: Models inspired by the human brain that learn tasks from data.

- Probabilistic
- Machine learning
- Data-driven

Machine learning

What is machine learning?

Machine learning is a branch of artificial intelligence that enables machines to learn from data without being explicitly programmed to perform specific tasks.

- **Discriminative AI:** Models the boundary between classes.
  Examples: Logistic regression, SVMs and neural networks for classification.

- **Generative AI:** Models the data distribution to generate new data.
  Examples: GANs, VAEs and language models such as GPT.

Discriminative machine learning

Machine learning is a branch of artificial intelligence that enables machines to learn from data without being explicitly programmed to perform specific tasks.

Machine learning encompasses two broad types of learning:

- **Supervised:** Algorithms learn from labelled data to make predictions or classifications.
- **Unsupervised:** Learning uses unlabelled data to discover hidden structures.

Supervised learning

Supervised learning involves learning a model that maps a set of *features* to a *label*.

- **Inputs:** A labelled dataset used to train the model.
  Example: Texts (tweets, etc.) labelled with positive or negative sentiment.
- **Output:** A label for a data point the model has not seen.

Supervised learning itself has two main types:

- **Classification:** Predicting a category or class.
  Examples: An image label (cat, dog, etc.), the sentiment of a text, or a customer's interests inferred from their comments.
- **Regression:** Predicting a continuous value, typically a real number.
  Examples: The price of an apartment or a customer's lifetime value.

Classification

Classification example: predictive maintenance

Age (years)	Temp. (°C)	Vibration (mm/s)	Hours/ week	Failures	Maintenance	Label (y)
2	55	1.2	40	0	Yes	No failure
6	70	3.8	60	2	No	Failure
4	65	2.5	50	1	Yes	No failure
8	80	4.1	55	3	No	Failure
3	58	1.9	45	0	Yes	No failure

Regression: daily volume dispensed at a fuel depot

Day	Temp. (°C)	Price (€/L)	Promo (0/1)	Stock remaining (%)	Volume dispensed (m³/day)
Mon	23	1.98	0	65	380
Tue	25	1.95	1	60	420
Wed	22	1.96	0	58	360
Thu	30	1.94	1	55	500
Fri	28	1.97	0	52	470
Sat	26	1.99	0	50	430
Sun	20	2.01	0	70	280

Unsupervised learning

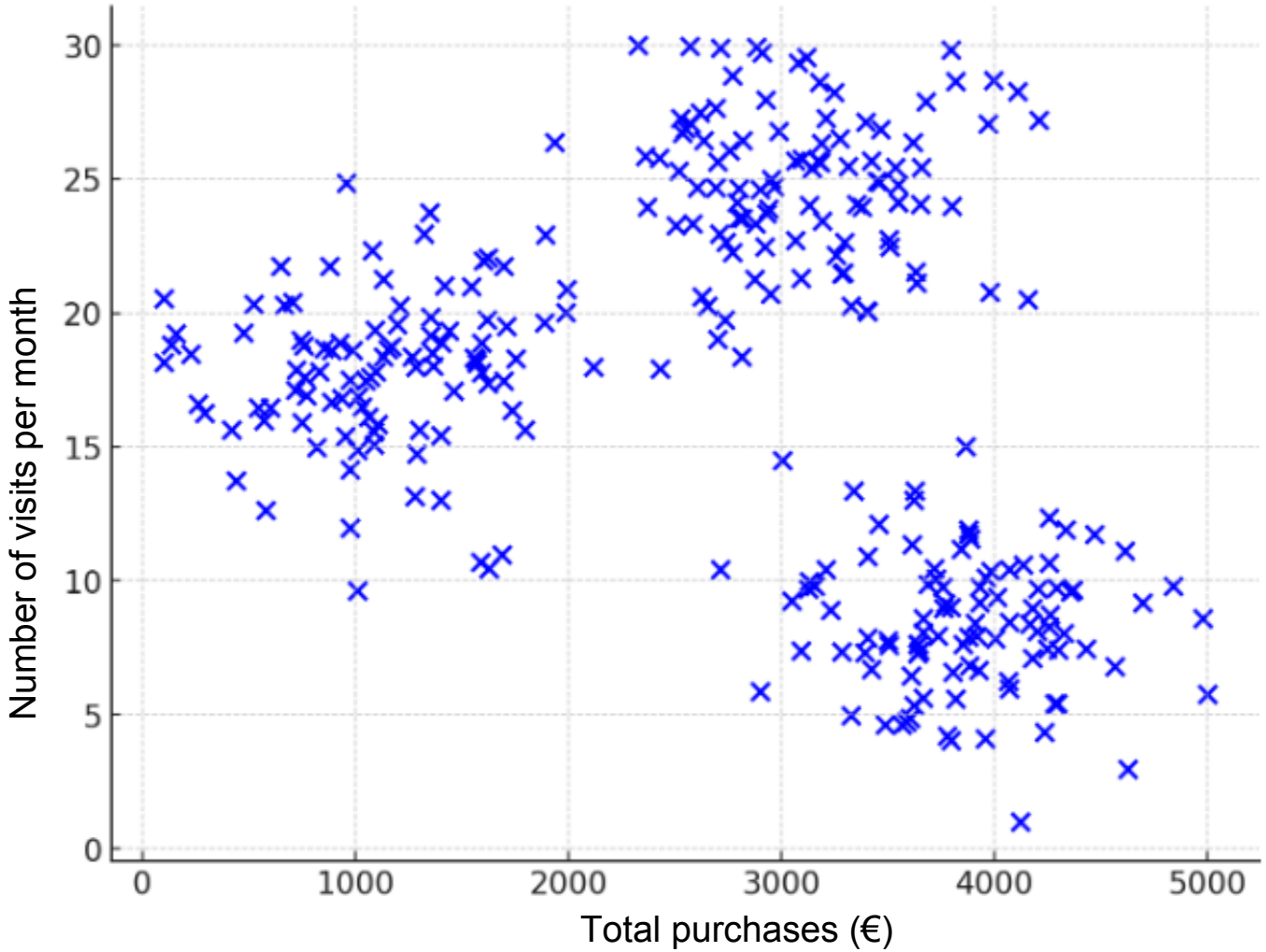
Unsupervised learning uses machine learning models to identify patterns and structures in unlabelled data.

Main types of unsupervised learning:

- **Clustering:** Grouping similar data points together. Examples: Market segmentation and social grouping.
- **Anomaly detection:** Detecting observations whose characteristics are unusual compared with the majority.

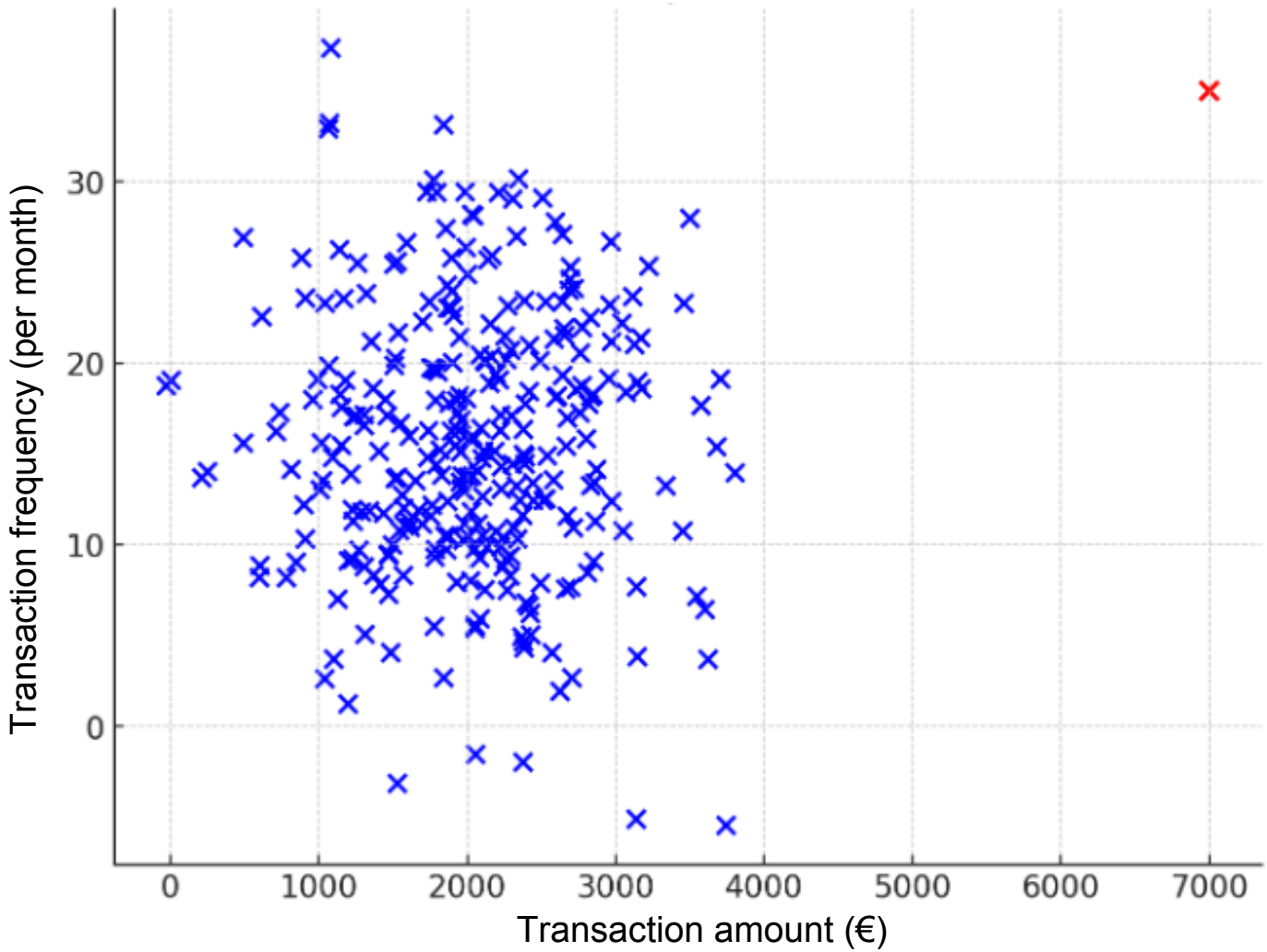
Clustering

Clustering example: customer segmentation



Anomaly detection

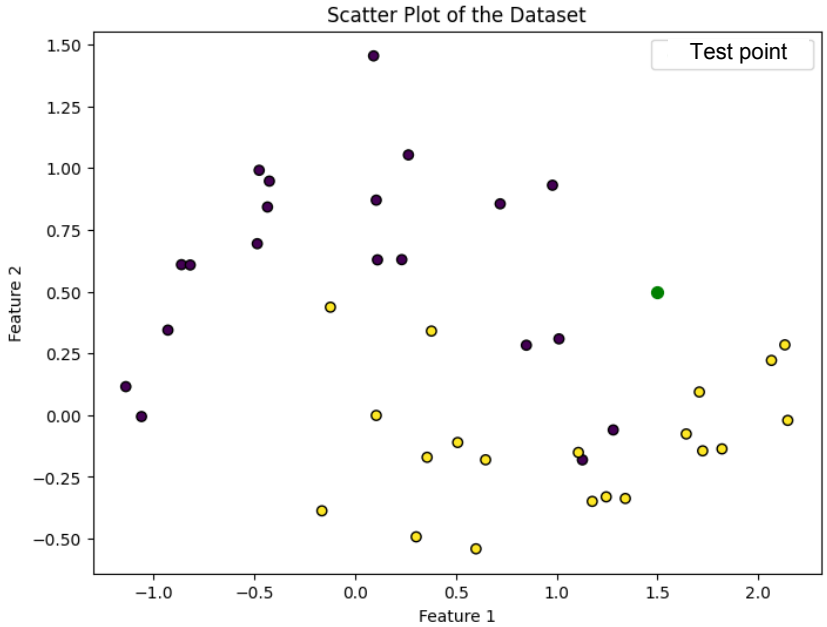
Anomaly detection example: bank fraud



Supervised learning

Supervised learning as an induction problem

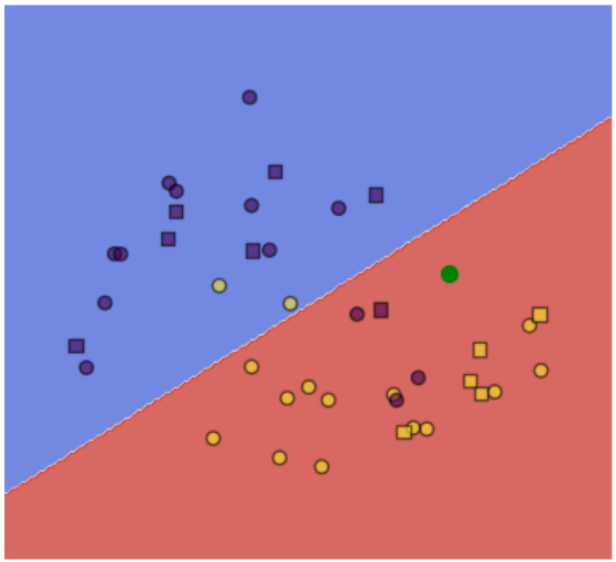
- **Definition:** Supervised learning learns a function *f* mapping inputs *X* to outputs *y*, from training examples (*X*, *y*).
- **Induction:** The model infers a general rule from specific data, with the aim of generalizing to new instances.
- **The generalization problem:** How can we ensure that the learned rule applies to new data?



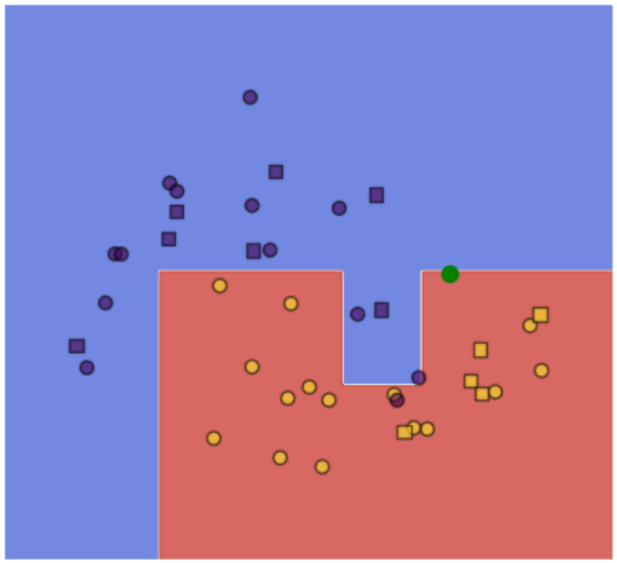
Model selection is inherently underdetermined

There are infinitely many ways to infer a model from a sample of training data.

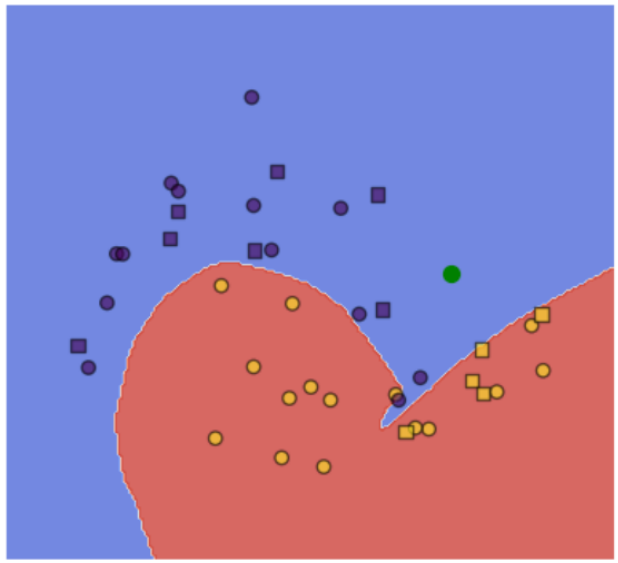
Logistic regression (n=40)



Decision tree (n=40)

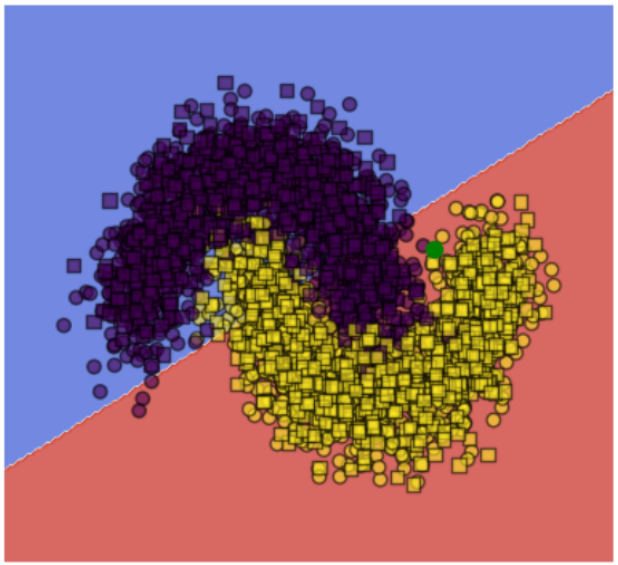


Deep neural network (n=40)

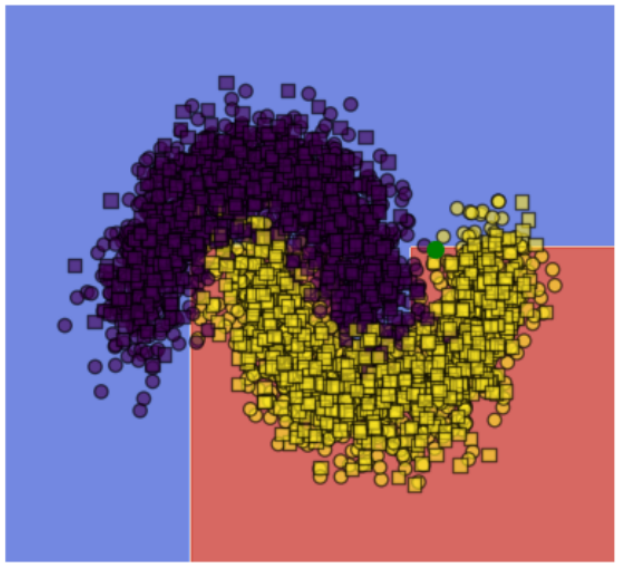


Underfitting and overfitting the training data

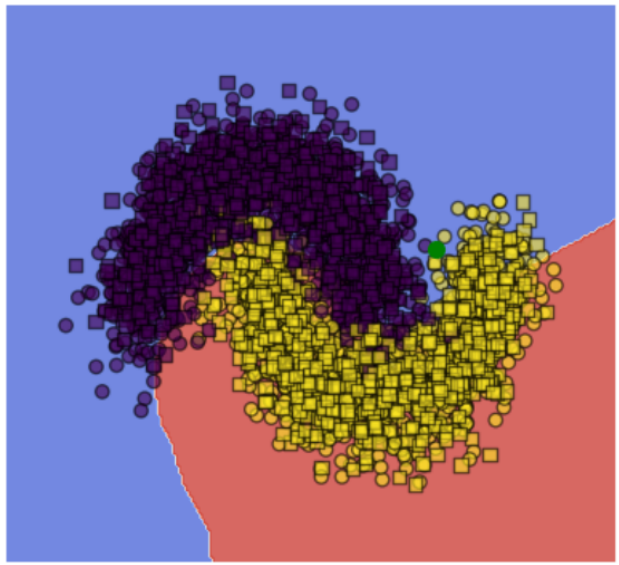
Logistic regression (n=3000)



Decision tree (n=3000)



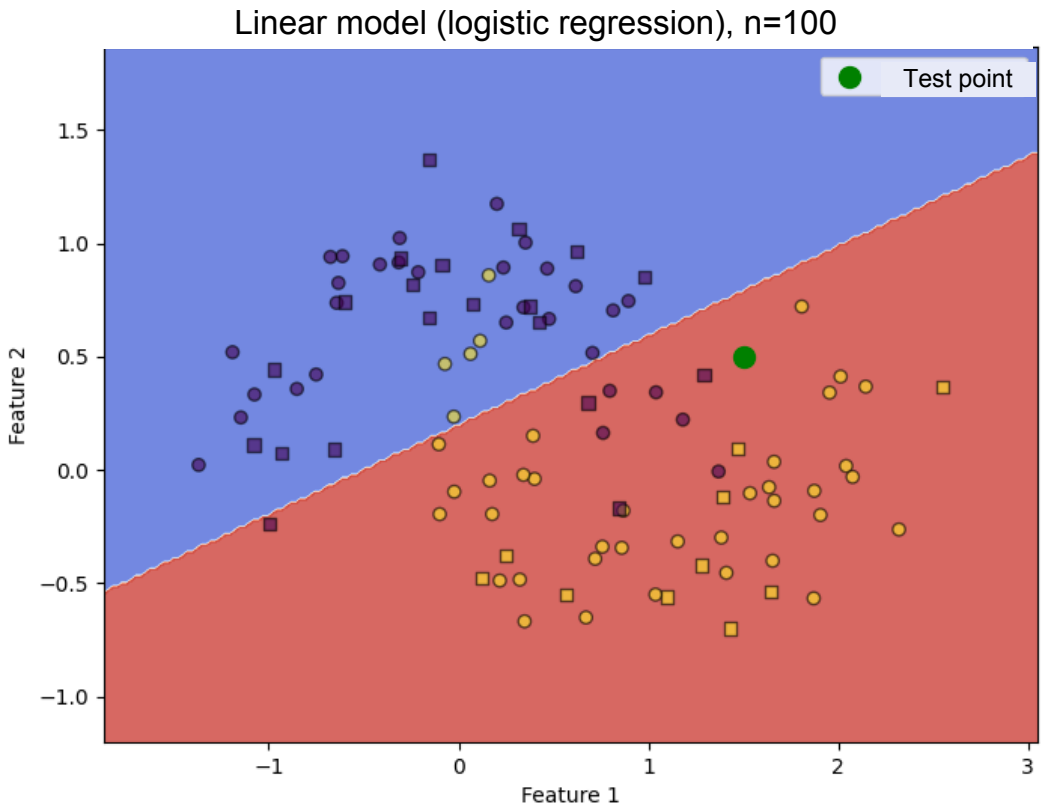
Deep neural network (n=3000)



Underfitting

Definition

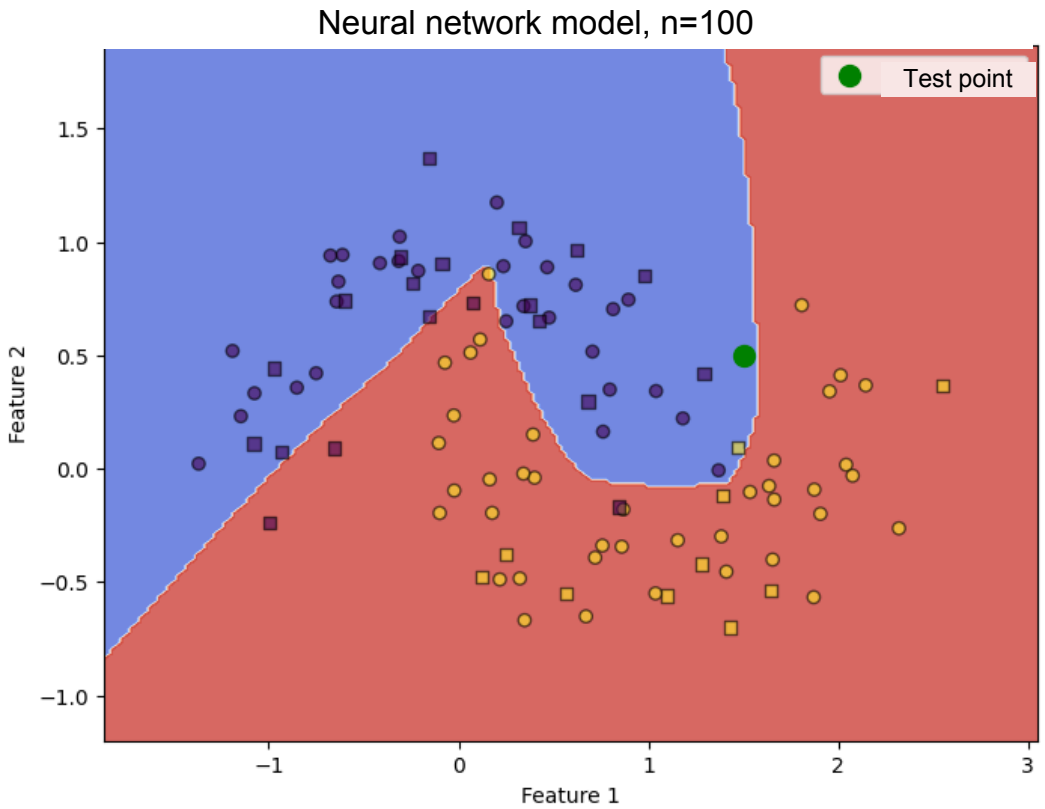
A machine learning model underfits when it fails to capture the complexity (information) present in the training dataset.



Overfitting

Definition

A machine learning model overfits when it fails to generalize to unseen data, i.e. when it fits the training data too closely.



Model selection

To select the most suitable model for a given metric, use the following procedure:

- Split the available dataset into three parts: a training set, a validation set and a test set.
- Train *M* models on the training set.
- Evaluate the *M* models on the validation set and select the best one.
- Evaluate the selected model on the test set. Ideally, the test set is used only once.



Example: model selection

Model	Training accuracy	Validation accuracy
Logistic regression	0.90	0.88
Decision tree	0.95	0.92
Deep neural network	1	0.90

Table: Comparison of model accuracies on the training and validation sets.

Performance metrics: regression

Several metrics can be used to evaluate machine learning models. These fall into two categories.

Regression

• **Mean squared error (MSE):** The average squared difference between predictions and observed values.

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^{n} (f(x_i) - y_i)^2$$

• **Root mean squared error (RMSE):**

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (f(x_i) - y_i)^2}$$

Performance metrics: classification 1/2

Accuracy: A basic metric for evaluating a classification model, defined as:

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{Total number of predictions}}$$

Confusion matrix: Provides more detail than accuracy, especially for imbalanced datasets (with majority classes). It compares model predictions with actual values and is structured as follows:

	Predicted positive	Predicted negative
Actual positive	True positive (TP)	False negative (FN)
Actual negative	False positive (FP)	True negative (TN)

Performance metrics: classification 2/2

Other metrics can be derived from the confusion matrix:

- **Precision:** The proportion of correct predictions among all positive predictions.

$$\text{Precision} = \frac{TP}{TP + FP}$$

- **Recall:** The proportion of actual positives correctly identified by the model.

$$\text{Recall} = \frac{TP}{TP + FN}$$

- **F1-score:** The harmonic mean of precision and recall.

$$\text{F1-score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$